

GEYI YANG

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Shenzhen, Guangdong, China

RESEARCH INTERESTS

GUI agents, self-improving agentic systems, harness optimization, and retrieval-augmented generation (RAG).

EDUCATION

- **The Chinese University of Hong Kong, Shenzhen** Sep. 2023 – Present
B.Eng. in Computer Engineering Shenzhen, China
 - GPA: 3.73/4.00.
 - Selected coursework: Artificial Intelligence, Machine Learning, Optimization, Data Structures, Operating Systems, and Internet Architecture and Protocol.
- **University of California, Berkeley** Jan. 2026 – Jun. 2026
GLOBE Program Visiting Student in Computer Science Berkeley, CA, USA

HONORS AND AWARDS

- COMAP Mathematical Contest in Modeling (MCM), Finalist (F Award; top 1%) 2026
- China International College Students' Innovation Competition (CICSIC), Bronze Award 2025
- Dean's List, CUHK-Shenzhen 2023–2024, 2024–2025, 2025–2026
- Academic Performance Scholarship Awardee, CUHK-Shenzhen 2023–2024

RESEARCH EXPERIENCE

- **GUI-HarnessGrad: Automated Harness Optimization for GUI Agents** Jul. 2026 – Present
Advisor: [Prof. Zhongxiang Dai](#), CUHK-Shenzhen; work in progress for ICLR 2027
 - Develop an automated algorithm that improves a GUI agent's inference-time harness from rollout failures while keeping the underlying model fixed, rather than hand-designing one task-specific harness.
 - Built a closed loop of trace analysis, failure clustering, harness-component attribution, candidate updates, paired retesting, and retain-or-rollback decisions; optimize success rate under explicit step and cost budgets.
 - On OSWorld (361 tasks; 15-step budget), improved a fixed Qwen3-VL-32B-Instruct agent from 37.67% to 50.95% (+13.28 points; +35.2% relative), exceeding reported results for Claude 4 Sonnet, o3, and several specialized GUI-agent models.
- **Meme-Qwen (7B-Instruct)** Jan. 2026
Chinese Internet-Meme Conversational Language Model  Model |  Dataset
 - Independently built and open-sourced, to our knowledge, the first Chinese internet-meme conversational SFT dataset and the first open model of this type: collected 30,000+ post-comment pairs with Playwright and released 8,680 ChatML examples.
 - Fine-tuned Qwen2.5-7B on a single A100 (40GB) with LoRA and DPO; improved meme-use accuracy by 32%, with human and LLM preference rates of 82%/91%/68% for humor, accuracy, and naturalness.

INTERNSHIPS

- **Cloud Computing Center, Chinese Academy of Sciences** Mar. 2025 – Jun. 2025
Research Intern – Domain-Specific Agentic RAG for Mining-Blasting QA Dongguan, China
 - Built and optimized an Agentic RAG system for retrieving and answering questions over mining-blasting technical documents, owning the end-to-end retrieval and reasoning pipeline.
 - Architected a hierarchical LangGraph workflow for memory, query planning, answer synthesis, and a ReAct subgraph for tool use and multi-step reasoning; added human clarification for ambiguous queries.
 - Developed Docling-based parsing and parent-child hybrid retrieval in Qdrant, combining BGE-M3, BM25, and BGE-Reranker-v2-m3; achieved 81% Recall@5 and 78% answer accuracy on 300+ QA cases.
- **Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)** Sep. 2023 – Jun. 2024
Research Intern – Multimodal Digital Human Synthesis for Emotional Interaction Shenzhen, China
 - Developed an end-to-end “Digital Relative” pipeline from family photos and recordings, integrating MockingBird voice cloning, SadTalker lip-sync and facial animation, and ChatGPT-powered dialogue.
 - Piloted the system with family users in 500+ interactions: 83% reported a familiar emotional connection and 20% reported a strong emotional response.

SELECTED PROJECTS

- MCM 2026: Dynamic Hybrid Lunar Logistics Optimization**
Modeling and Programming Lead

2026
Finalist (F Award; top 1%)

 - Developed a dynamic Earth-to-Moon logistics model integrating space elevators, reusable rockets, and ISRU maturation; formulated time-cost objectives and operational constraints.
 - Used Differential Evolution (1,301 evaluations), Monte Carlo analysis (5,000 simulations), and sensitivity analysis to identify a robust three-phase transition strategy; awarded Finalist (top 1%).
- Telos: AI-Generated, Ready-to-Learn Courses**
Independent Full-Stack Developer

Aug. 2026 – Present

 - Building a goal-conditioned education product that turns user materials and cross-platform resources into structured, multimodal, ready-to-learn courses rather than isolated answers or learning plans.
 - Designed an agentic pipeline for retrieval, deduplication, quality assessment, sequencing, and embedding source text, images, and videos in a unified learning workspace.
- Voice Turing Test: WeChat Mini Program and Research Data Platform**
Full-Stack Development Lead

May 2025 – Aug. 2025

 - Independently led registration, UI/UX, full-stack development, testing, and deployment of a gamified Human-vs-AI speech challenge used as the data-collection platform for *Human or Machine?* (ICLR 2026).
 - Built the native WeChat front end and Tencent Cloud backend with bilingual UI, real-time scoring, achievements, leaderboards, and referral rewards; reached 1,000+ users, 300+ DAU, and 5,000+ valid records in the first month.

TEACHING EXPERIENCE

- The Chinese University of Hong Kong, Shenzhen**
Teaching Assistant – MAT1001 Calculus I

Sep. 2025 – Dec. 2025
Shenzhen, China

 - Served as a teaching assistant for the first-year calculus course MAT1001, supporting tutorials and student learning.

SKILLS AND LANGUAGES

Programming & Systems: Python, C++, JavaScript, TypeScript, Shell, FastAPI, React, SQLite, Git, Linux.
Machine Learning & Agents: PyTorch, Transformers, LoRA/PEFT, TRL (DPO), RAG, LangChain, LangGraph, GUI agents, Qdrant, MCP.
Languages: English (TOEFL 111; GRE 332).